Experimental Design & Scope of Inference

Section 1: Observational Studies vs. Experiments

1 What essential feature distinguishes a randomized experiment from an observational study?

a. Treatments are actively imposed on experimental units by the researcher

b. A larger sample size is used

c. Participants are randomly surveyed by phone

d. Data is collected over a longer time period

2 Researchers observe that people who drink green tea daily have lower rates of cardiovascular disease. Can they conclude that green tea prevents heart disease?

a. No, observational studies cannot establish cause-and-effect due to potential confounding variables

b. Yes, because the association is statistically significant

c. Yes, provided at least 10,000 subjects were observed

d. Only if green tea drinkers were matched by age

3 Only which type of statistical investigation allows researchers to establish cause-and-effect conclusions?

a. A well-designed randomized comparative experiment

b. A large-scale stratified observational study

c. A voluntary response poll with $n > 50,000$

d. A census of the population

Section 2: Principles of Experimental Design

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4 What is the primary purpose of random assignment in an experiment?

a. To create roughly equivalent treatment groups and balance out confounding variables

b. To ensure the sample represents the entire population

c. To eliminate all experimental error

d. To prevent the need for a control group

5 What role does a control group serve in an experiment?

a. Provides a baseline measurement to compare against active treatments

b. Ensures the experiment has a large enough sample size

c. Guarantees that subjects do not know which treatment they receive

d. Eliminates the need for replication

6 In experimental design, what does 'replication' mean?

a. Assigning multiple experimental units to each treatment group

b. Repeating the entire experiment in another country

c. Using identical equipment across all trials

d. Testing subjects multiple times in a row

Section 3: Blinding, Placebos, and Confounding

7 What is a double-blind experiment?

a. Neither the subjects nor the individuals evaluating the outcomes know which treatment each subject received

b. Neither the researcher nor the subjects know the purpose of the study

c. The experiment is conducted in a darkened laboratory

d. Both the control group and treatment group receive identical active medications

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8 What is the 'placebo effect'?

a. When subjects show improvement simply from the psychological expectation of receiving a treatment

b. When a drug causes adverse side effects in control patients

c. When experimental units drop out of a clinical trial

d. When researchers unconsciously bias results

9 What is a confounding variable?

a. A variable related to both the explanatory variable and response variable that obscures whether the treatment caused the outcome

b. A variable that is held constant across all treatment conditions

c. A calculation error during hypothesis testing

d. A categorical variable with more than two levels

Section 4: Blocking and Scope of Inference

10 When should an experimenter implement a randomized block design?

a. When there is an identified characteristic expected to affect the response, so subjects are grouped by that trait before randomization

b. When the sample size is too small for simple random assignment

c. When double-blinding is completely impossible

d. To avoid having to use a control group

11 A matched pairs design is a special form of:

a. Block design

b. Cluster sampling

c. Stratified survey

d. Completely randomized design

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12 If subjects are volunteers (not randomly selected from a population) but are randomly assigned to treatments, what inference can be drawn?

a. Causation can be inferred, but only for individuals similar to the volunteer participants

b. Results can be generalized to the entire national population

c. No valid statistical conclusions can be drawn

d. Association only, causation cannot be established

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